# Status Report

## Using Machine Learning to Predict Scattering Parameters for Signal Integrity

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# 1 Introduction and Problem Description

Signal integrity (SI) is getting more important for modern digital systems. As the data rates on a printed circuit board (PCB) rise, the interconnect performance degrades because the PCB structure - like multilayer stacks with vias, traces as well as other components - introduce signal loss, reflections as well as crosstalk and noise. These effects can be described using scattering parameters (S-parameters) [4].

Typically, simulation-based methods such as full-wave electromagnetic simulation are used to predict S-parameters for PCB interconnects. They are accurate, but time consuming, requiring extraordinary computation [5]. This is the motivation of the project: using machine learning (ML) techniques to provide alternative methods.

By making use of SI/PI-Database provided by Institut für Theoretische Elektrotechnik [1], this project trains ML models to computationally effectively predict the scattering parameters of PCB interconnects.  Physically meaningful constraints would be considered [2] while maintaining acceptable accuracy and physical consistency.

# 2 Background and Literature Review

S-parameters describe how electromagnetic waves propagate through a network. It is commonly used by modern engineers to analyse the attenuation, reflections, crosstalk as well as other effects of signal propagation on PCBs [4], [5].

Most traditional methods are physics-based (PB) [9], [10], such as full-wave electromagnetic simulations including the finite element method (FEM) and finite difference time domain (FDTD) [1]. For complex PCB structures, these numerical techniques are accurate but time-consuming especially when many design variations are required [5]. As a result, they are becoming bottlenecks of modern high-speed electronic system developments, as the paces of design-implementation-optimization loops keep increasing.

Recently, ML technologies have been rapidly evolving, and widely adopted by different industries in different areas including image processing, voice recognizing, content generating, etc. Generative AI are very popular nowadays, like ChatGPT from OpenAI, Gemini from Google, Claude from Anthropic, etc., and they are all built on top of ML technologies. This proves ML technologies have been well-developed.

However, nowadays, ML tools and technologies are not easily adopted in electronical engineering applications, especially because of requirements of physical consistency [11]. Another important aspect to adopt ML in electronical engineering is dataset. Unlike IT industry where data is commonly shared, like ImageNet and OpenImages [12], [13], PCB simulation data sharing is not as common in the electronical engineering community. Fortunately, the import of open-source datasets is increasingly recognized. For example, Institut für Theoretische Elektrotechnik developed has open-sourced the SI/PI-Database [1] project, and European Open Science Cloud provides the collection of research data [14].

With well-developed ML technique and simulation dataset available, it is now feasible to train ML models as data-driven alternative for electromagnetic modelling. By feeding PCB structure design parameters and the corresponding electromagnetic responses derived from the Touchstone files, ML models can discover the underlying patterns between them. Thus, ML models can be used as surrogate models that make prediction like traditional simulation methods do, but in a significantly faster way. [3], [7].

As a type of ML techniques, neural networks have been widely used to model complex, nonlinear relationships between the input, that are PCB structure parameters, and the output electromagnetic responses [7], [8]. Different studies have shown that neural networks have the capability to predict S-parameters or insertion loss (IL) on high-speed interconnects, and yield reasonable accuracy. This shows the potential to adopt neural networks to accelerate signal integrity analysis [2], [3].

On the other hand, studies also showed that an adaption to specific requirements of signal integrity is needed [1], because purely data-driven neural network models can make predictions that not physically reasonable. In electromagnetic modelling, physical constraints must be complied, such as insertion loss must be a non-negative quantity. Fortunately, recent work has explore the feasibility to use physics-enforced neural networks to address this problem by incorporating physical constraints to the modelling design [2][6].

Based on these researches and open-source datasets, this project explores the use of ML tools and techniques as a computationally efficient alternative to predict S-parameters for PCB interconnect structures for the purpose of signal integrity. The intended models should be reasonably accurate while maintaining physical consistency.

# 3 Work Completed to Date

**1. Problem framing and literature reviews.** Several papers have been reviewed, provided provisional understanding on the ML approaches for SI/PI modelling and the potential methods to enforce physical consistency in electromagnetic engineering [1][2]. Especially, the SI/PI database project was studied to gain knowledge of PCB structures, terminologies, design parameters and the Touchstone file formats[1]. Several ML methods in SI/PI packaging, as well as model-based design paradigms, were also reviewed to explore possible model choice and validation approach [3].

**2. Project goal clarification and model input/output definition.** After the researches and reviews, the project goal have been iteratively narrowing down from *Use of Machine Learning to Predict Circuit and System Behaviour in the Presence of Manufacturing Variability* to *Using Machine Learning to Predict Scattering Parameters for Signal Integrity*. This ensure the feasibility to development a meaningful project. The PCB interconnect structure parameters (including geometric and material parameters such as via pitch, via radius, cavity height, etc, alone with frequency) were selected as model inputs, and the output were the predictions of S-parameters. Candidate outputs were defined as (a) scalar SI metrics and (b) frequency-dependent responses extracted from Touchstone data.

**3. Dataset preparation and pipeline drafting.** Relevant open-source database projects were explored, including CircuitNet project [15] and SI/PI Database project [1]. The initial pipeline was also designed to load and vectorize PCB design parameter, build ML models, predict S-parameters and compare with ground true response from Touchstone files [1].

**4. Initial dataset analysis and interpretation.** Example datasets were also downloaded and preliminarily processed and analysed. By using data analysis techniques, the datasets were processed and plotted to validate data sanity and discover important parameters and their correlations and expected frequency-dependent trends.

**5. Formulation roadmap established (Formulations 1–4).** These modelling formulations were planned to progressively iterate from simple to complete: (1) non-neural regression model as a baseline for further dataset exploration and as a starting point; (2) scalar regression model to explore adopting neural networks; (3) multi-output regression neural network model predicting the full insertion-loss curve over discrete frequency points; and (4) full S-matrix prediction as a stretch goal. Due to the balance of SI usefulness and complexity, formulation 3 was selected as the intended final deliverable.

# 4 Proposed Solution

## 4.1 Overall roadmap

After preliminarily sanity checks,  the SI/PI database was selected as primary data source. It provides geometric and material parameters per design instance, as well as corresponding Touchstone S-parameter responses over frequency [1]. The objective is to train a model to predict SI-relevant target response from design parameters [2], [3].

## 4.2 Progressive modelling formulations

The work is organized into four increasingly complete formulations, to ensure a measurable progress and reduce deliverable risks:

**Formulation 1 - Baseline non-neural regression.** Design and train a traditional non-linear ML model apart from neural networks, such as tree-based or kernel regression, on low-dimensional targets to validate preprocessing, feature scaling, as well as evaluation pipelines.

**Formulation 2 - Scalar SI metric prediction.** Shift the modelling to a simple neural network model to predict a single scalar metric derived from a frequency response, such as insertion loss at a representative.

$$Y_k = IL_{mean}(10\,\mathrm{GHz})$$

**Formulation 3 - Multi-output regression (target formulation).** Iterate the modelling to predict the full insertion-loss curve at discrete frequency points for each design instance $k$:

$$Y_k = [\, IL(f_1),\, IL(f_2),\, \dots,\, IL(f_{200})\,]$$

This is the intended final deliverable. It retains frequency-dependent behaviour and directly supports SI comparisons across designs.

**Formulation 4 - Full S-parameter matrix prediction.** This is a optional stretch goal. Extend prediction to the full frequency-dependent S-parameter matrix $S_{ij}(f)$. The model introduces stronger physical constraints , but on the other hand would increases output dimensionality leading it much harder to be fine trained [3]. This will be attempted only if time permits after Formulation 3 is stable.

## 4.3 Target definition and physical consistency

Preliminarily, the insertion loss (IL), which is derived from transmission magnitude, was selected to play a key SI metric:

$$IL = -20 \log_{10} |S_{21}|$$

To ensure physical consistency, physics-enforced mechanisms informed by prior work will be incorporated in the model [2]. Cross-talk, eye-diagram and other potential metrics would be considered if needed during the project development.

## 4.4 Training and evaluation plan

**Data preparation.** Under the permit of SI/PI Database team, two of datasets were already downloaded. For each design instance, from formulation 1 to 4 aforementioned, the simulation dataset will be split into training, validation and testing sets to an appropriate proportion. Feature engineering will be required to select suitable geometric and material parameters as inputs, and the output will be derived from corresponding Touchstone files.

**Model training.** The aforementioned formulated models were design to be supervised regression models. For each design instance, the train progress will be in three stages: (1) build a trainable model that could learn meaningful data representation; (2) fine tune the model to have the capability to overfit; (3) improve the generalisation of the model [16].

**Evaluation metrics.** In preliminary design, performance of models will be assessed using mean squared error (MSE) and curve-level error measures over frequency, such as average error over all frequency points, and band-limited errors in SI-relevant frequency ranges. Additional checking method will be developed to evaluate physical consistency [2].

# 5 Project Schedule and Success Criteria

## 5.1 Project plan and Schedule

To align with the progressive modelling formulations, this project will be developed in several staged milestones. The dissertation thesis will be written during the milestones.

- **Milestone 1 - Data pipeline completed (Weeks 1–2).**

**Task:** (1) finalise dataset selection; (2) implement parsing of parameter.csv and Touchstone files; (3) define train/validation/test split; (4) implement normalisation and target extraction.

**Deliverables:** reproducible data-preparation scripts; sanity-check analysis and plots.

**Milestone:** further understand the problem and dataset, preprocessing training tensors.

- **Milestone 2 - Baselines and scalar targets (Weeks 3–4).**

**Task:** (1) train baseline non-neural regressors (Formulation 1); (2) train scalar regression models (Formulation 2); (3) select evaluation metrics;(4) establish baseline performance.

**Deliverables:** baseline results table and error plots; brief model comparison.

**Milestone:** baseline model predicts scalar SI metric with reasonable generalisation.

- **Milestone 3 - Multi-output curve prediction (Weeks 5–7).**

**Task:** (1) implement Formulation 3 - multi-output regression for full insertion-loss curve prediction ; (2) tune model architecture and hyperparameters.

**Deliverables:** trainable Formulation 3 model and generalisation evaluation.

**Milestone:** stable prediction with acceptable accuracy and generalisation on unseen data.

- **Milestone 4 - Physical consistency enforcement (Weeks 8–9).**

**Task:** incorporate physical consistency mechanisms;

**Deliverables:** physically constrained model variant; ablation study showing impact on accuracy vs. physical validity.

**Milestone:** model satisfies key physical checks.

- **Milestone 5 - Final evaluation and reporting (Weeks 10–12).**

**Task:** (1) consolidate experiments; (2) document methodology; (3) prepare final report and presentation materials; (4) (optional) attempt extension toward Formulation 4 if time permits.

**Deliverables:** final dissertation artefacts; reproducible code and results; presentation slides.

## 5.2 Key Risks / Issues and Mitigations

- **High output dimensionality:** may cause unstable training or overfitting.
  **Mitigation:** (1) progressive approach; (2) regularisation and validation.
- **Non-physical predictions:** models may violate physical constraints.
  **Mitigation:** incorporation of physics-enforced mechanisms.
- **Dataset limitations:** dataset size/coverage may limit generalisation.
  **Mitigation:** robust train/test split, error analysis across parameter ranges.

## 5.3 Success Criteria

The project will be considered successful if:

1. Intended model predicts the target metrics derived from ground true Touchstone result for unseen test samples with consistent generalisation and reasonable accuracy.

2. Intended model outperforms baseline model (Formulation 1).

3. The final model demonstrates physically meaningful behaviour.
4. Results include a clear evaluation report: quantitative errors, representative predicted compare to ground-truth curves, and discussion of limitations and applicability for SI purposes.

# 6 Declaration of GenAI Use (TOPIC)

## 6.1 TOOLS

1. Tool: ChatGPT (OpenAI)
2. Model: CPT-5.2 Thinking
3. Date of use: March 2026

## 6.2 OUTPUTS

1. Summaries of referenced papers
2. Refine citations of referenced papers
3. How to use Microsoft Word for editing
4. Checkup electromagnetic engineering domain terminologies
5. LaTeX syntax checkup

## 6.3 PROMPTS

1. Summarize referenced paper xxx.pdf
2. Refine the citations sections
3. How to insert an equation in Microsoft Word
4. How to show page numbers in Microsoft Word
5. How to setup content table in Microsoft Word
6. Is "via" (and other terminologies) a term of electronic engineer?
7. Convert the equation to LaTeX syntax

## 6.4 ITERATIONS

The interaction was iterative. Several sections were repeatedly revised by prompts like:

1. Use IEEE-style in-text citations
2. Shorten content to meet the five-page limit

## 6.5 CRITICAL REFLECTION

The GenAI was treated as assistant tool to checkup information, assist materials reading like prior study works, rather than authoritative sources. The tool was useful for editor (like Microsoft Word) usage, improving clarity and compressing text to fit the page limit. To avoid over-general statements or inaccurate references, therefor:

1. The information provided by the tool was cross-checked against sources used in report.
2. References suggested by the tool were verified and corrected where necessary.
3. All decisions on the project scope, formulation and success criteria were made by the author under the help of the supervisor.

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